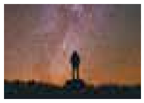
NASA exoplanet exploration
Learn more about the NASA exoplanet exploration program to find life beyond our solar system. **http://dgit.in/oct20-45**



Solutions to the Fermi Paradox
These are some possible solutions to the Fermi Paradox that you may find interesting. **http://dgit.in/oct20-46**

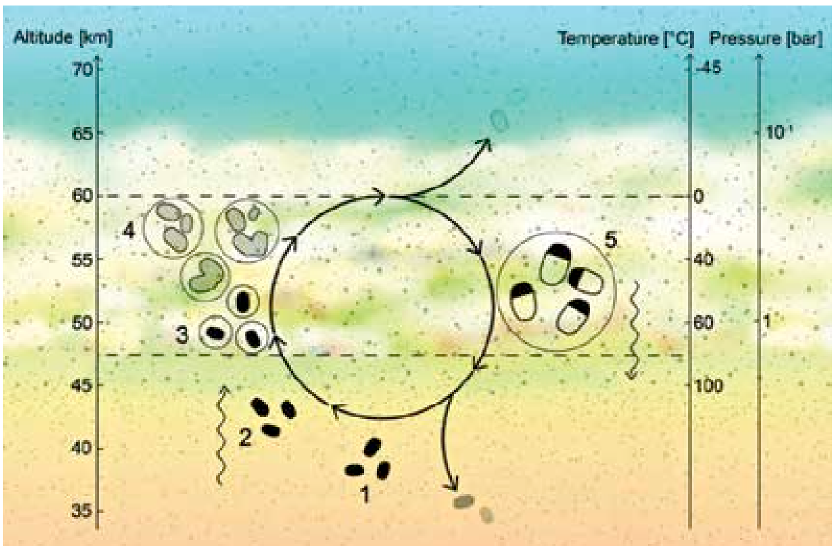
phosphine on Venus was made using two ground based telescopes, the James Clerk Maxwell Telescope (JCMT) in Hawaii and the Atacama Large Millimeter Array (ALMA) in Chile. Following the discovery, the researchers from MIT looked at all the possible ways in which phosphine could be introduced to the atmosphere, and eliminated all abiotic processes to create it. The conclusion was that as far as we know it, only life can create phosphine, there is phosphine on Venus, which indicates that there is life hiding in the atmosphere of Venus. Clara Sousa-Silva, a researcher at MIT says, “Now, astronomers will think of all the ways to justify phosphine without life, and I welcome that. Please do, because we are at the end of our possibilities to show abiotic processes that can make phosphine.”

Jane Greaves and team of Cardiff University first used the JCMT to identify patterns in light that could indicate the presence of biomolecules. In these searches, there were indications of the presence of Venus. Greaves then got in touch with Sousa-Silva, who was an atmospheric scientist with expertise in phosphine. Follow up investigations using the ALMA observatory confirmed the initial finding, showing that the characteristics of light from Venus were consistent with the presence of phosphine. The MIT team then used atmospheric models of Venus to probe any geological or abiotic causes that could introduce phosphine into the atmosphere. Researchers from Cambridge investigated how such processes could build up phosphine in the atmosphere. The finding was that the amount of phosphine detected in the atmosphere simply could not come from abiotic sources. While only 20 of every billion molecules in the atmosphere of Venus is phosphine, the concentration is more than the atmosphere on Earth.

The finding, however, is not altogether unexpected. In the 60s, Carl Sagan published a pair of papers that first showed that Venus had all the ingredients necessary for life, namely water, carbon dioxide and sufficient sunlight, and another which proposed that microorganisms could exist in the atmosphere of Venus. In the late 90s, a flurry of papers investigated the potential for the habitability of Venus. The characteristic yellow colour of Venus cannot be completely explained by its known atmospheric composition alone, even after factoring ice particles and aerosols. One possible explanation is that microbial life in the atmosphere may give Venus its colour, and what is more, telescopes have identified particles in the atmosphere that are consistent in size with microbial life. Over the years, probes and telescopes have identified other biomarkers, such as chlorine and carbonyl sulfide in the Venusian atmosphere, but these can also be produced by geological processes. Phosphine is the first biomarker to be identified that cannot be produced by any known geological source.

The team is now looking to investigate the layer in the atmosphere, as well as how it interacts with lower cloud layers. Follow up studies will characterise the distribution of phosphine, and find out if the concentration of phosphine shows any seasonal variations, and if there are any cycles.

Researchers have not directly observed or identified any life forms yet. Actually that is not entirely true, strange shapes have been observed on the images captured by the Venera spacecraft on the surface, called Amisadas. These objects cast fuzzy shadows unlike the hard shadows of rocks, appeared to move between photos and appeared to have distinct anatomical features. These photos are not conclusive. However, on a theoretical level it is possible that the unicellular cousins of Amisadas or other terrestrial organisms on Venus were launched into the atmosphere by volcanic activity. Researchers have also theoretically shown different mechanisms through which atmospheric microbes on Venus can stay suspended, metabolise, and give rise to the distinct coloration of Venus. Even before the discovery of phosphine, there were plans to study the atmosphere of Venus for signs of life. These concept missions are likely to get fast tracked following the discovery of phosphine on Venus. This month itself, BepiColombo is scheduled for the first of its two Venus flybys, and on board are instruments that might be sensitive enough to detect phosphine. The timing of the spacecraft flyby and the discovery of phosphine is just an astronomical coincidence.



Credit: Seger et al.

How microbes stay suspended in the atmosphere of Venus. Dehydrated microbes (1) are suspended in a layer of haze below the clouds, where updrafts (2) carry them upwards to the habitable cloud layer where they are hydrated (3) where the organisms can metabolise and multiply (4) forming large droplets that sink to the lower haze layer and are consequently dehydrated (5).